

Sequencing types

- Whole genome sequencing (human, plant, bacteria, etc.)
- Single cell sequencing
- Metagenomic sequencing
- RNA-Seq
- etc.

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Single cell sequencing

- Genetic variation between different tissues

Extensive genetic variation in somatic human tissues

Maeve O'Huallachain^{a,b}, Konrad J. Karczewski^{a,c}, Sherman M. Weissman^{d,1}, Alexander Eckehart Urban^{a,e}, and Michael P. Snyder^{a,1}

Mosaic Copy Number Variation in Human Neurons

Michael J. McConnell^{1,2,7,8,9}, Michael R. Lindberg⁷, Kristen J. Brennand^{1*}, Julia C. Piper^{1,2,†}, Thierry Voet^{1,4}, Chris Cowing-Zitron⁷, Svetlana Shumilina⁷, Roger S. Lasken^{5,6}, Joris R. Vermeesch⁷, Ira M. Hall^{1,9,‡}, Fred H. Gage^{7,‡}

- Genetic variation between healthy and diseased cells

Tumour evolution inferred by single-cell sequencing

Nicholas Navin^{1,2}, Jude Kendall¹, Jennifer Troge¹, Peter Andrews¹, Linda Rodgers¹, Jeanne McIndoo¹, Kerry Cook¹, Asya Stepanisky¹, Dan Levy¹, Diane Esposito¹, Lakshmi Muthuswamy¹, Alex Krasnitz¹, W. Richard McCombie¹, James Hicks¹ & Michael Wigler¹

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Single cell sequencing

- Sequencing of uncultured organisms

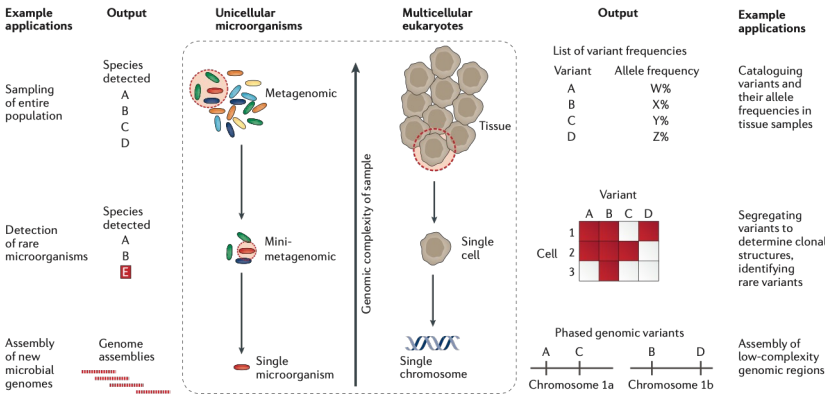
Insights into the phylogeny and coding potential of microbial dark matter

Christian Rinke¹, Patrick Schwientek¹, Alexander Sczyrba^{1,2}, Natalia N. Ivanova¹, Iain J. Anderson^{1,t}, Jan-Fang Cheng¹, Aaron Darling^{3,4}, Stephanie Malfatti¹, Brandon K. Swan⁵, Esther A. Gies⁶, Jeremy A. Dodsworth⁷, Brian P. Hedlund², George Tsiamis⁸, Stefan M. Sievert⁹, Wen-Tso Liu¹⁰, Jonathan A. Eisen¹, Steven J. Hallam⁶, Nikos C. Kyrpides¹, Ramunas Stepanauskas⁵, Edward M. Rubin¹, Philip Hugenholtz¹ & Tanja Woyke¹

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SINGLE-CELL OMICS

Single-cell genome sequencing: current state of the science



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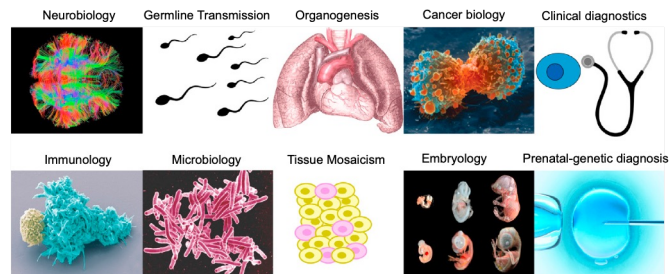
Advances and Applications of Single-Cell Sequencing Technologies

Yong Wang¹ and Nicholas E. Navin^{1,2,*}

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University of Texas, MD Anderson Cancer Center, Houston, TX 77030, USA

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<http://dx.doi.org/10.1016/j.molcel.2015.05.005>



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How does it work?

What are the biggest hurdles in Single Cell Sequencing?

- Cell isolation
- Whole Genome Amplification
- Data analysis

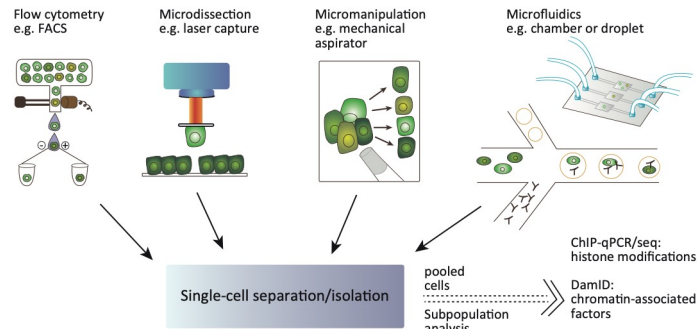
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Cell Isolation

Epigenetics reloaded: the single-cell revolution

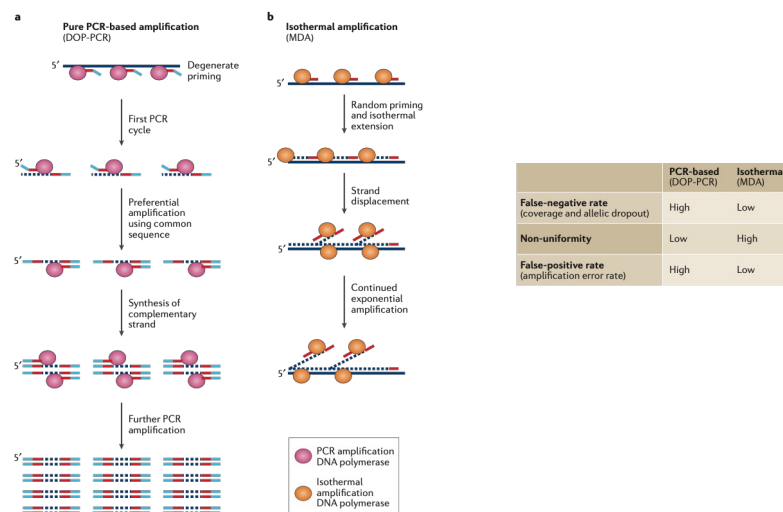
Poonam Bheda and Robert Schneider

Institut de Génétique et de Biologie Moléculaire et Cellulaire (IGBMC), CNRS UMR 7104/Inserm U984/Université de Strasbourg, 67400 Illkirch, France



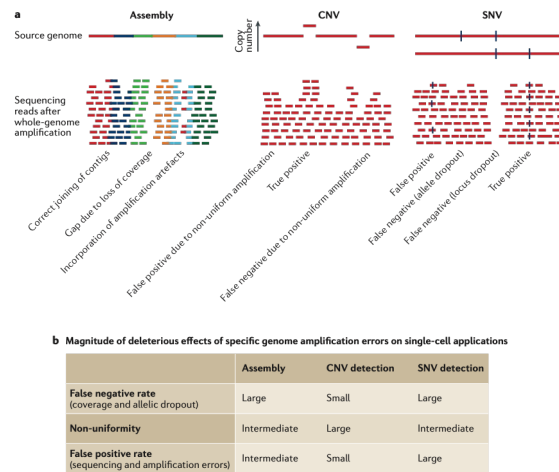
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Whole Genome Amplification



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Amplification and Sequencing bias



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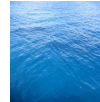
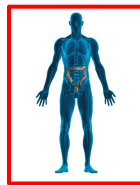
Metagenomics

- *META* = Beyond; transcending; more comprehensive
- It is the sequencing of DNA recovered directly from environmental or clinical samples

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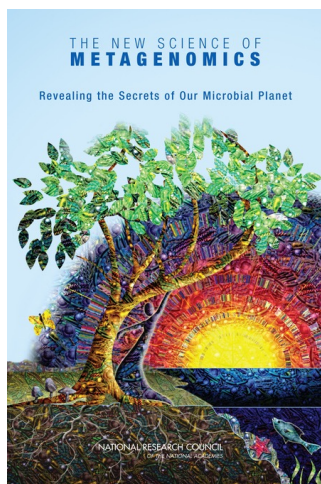
Q: What do these pictures have in common?

A: MICROBIOME
(a microbial community)



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Why study Microbiomes

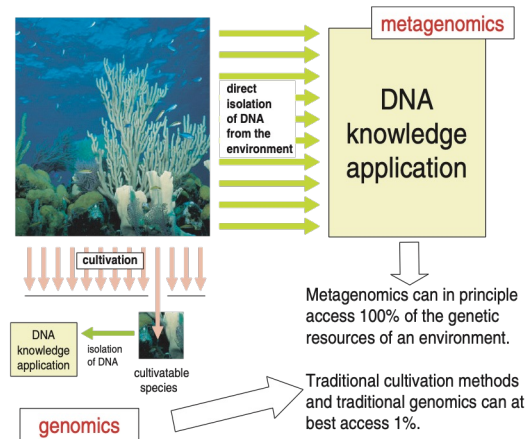


THE DAWNING OF A NEW MICROBIAL AGE

Microbes run the world. It's that simple. Although we cannot usually see them, microbes are essential for every part of human life—indeed all life on Earth. Every process in the biosphere is touched by the seemingly endless capacity of microbes to transform the world around them. It is microbes that convert the key elements of life—carbon, nitrogen, oxygen, and sulfur—into forms accessible to all other living things. For example, although plants tend to get credit for photosynthesis, it is in fact microbes that contribute most of the photosynthetic capacity to the planet. All plants and animals have closely associated microbial communities that make necessary nutrients, metals, and vitamins available to their hosts.

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Why study Microbiomes using Metagenomics?



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Importance of Microbiome Studies

Reef Health
and Bleaching



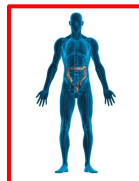
Disease
Transmission



Crop Health



Health,
Nutrition,
Diagnostics,
and Forensics



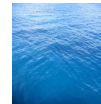
Basic Science
Synthetic Biology
Drug Discovery

Agriculture
And Detoxification

Geochemistry,
Carbon Cycle, and
Global Warming

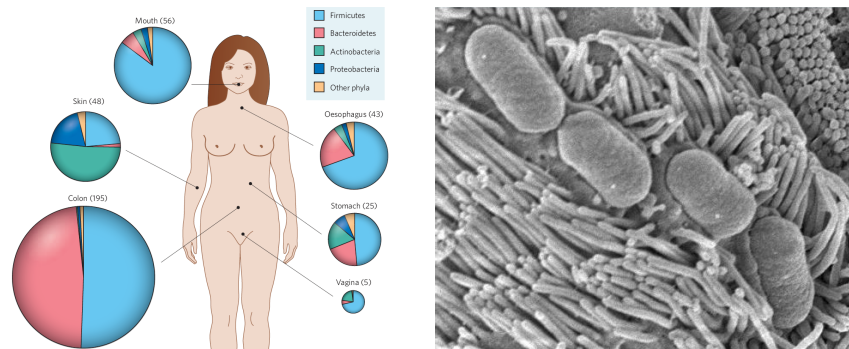


Water Pollution
and Toxicity



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The Human Microbiome



Adult human: 10^{13} mammalian cells + 10^{14} microbial cells

>1000 bacterial phylotypes

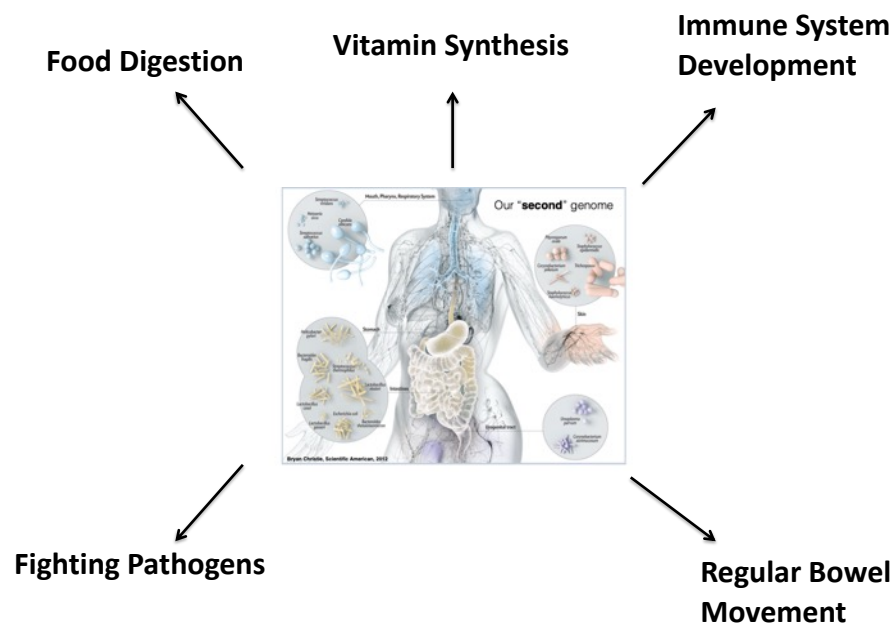
Human genome: 20,000 genes

Human supra-genome: 100 X human genes?

Sonnenburg *et al*, *Science* (2005) **307**, 1955
 Dethlefsen *et al*, *Nature* (2007) **449**, 811

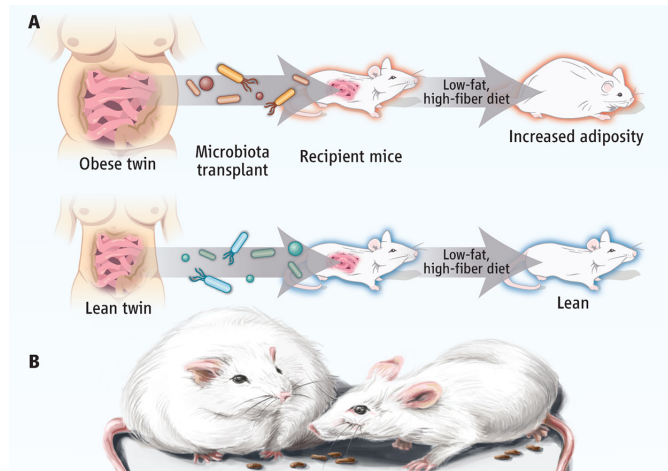
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Our Second Genome



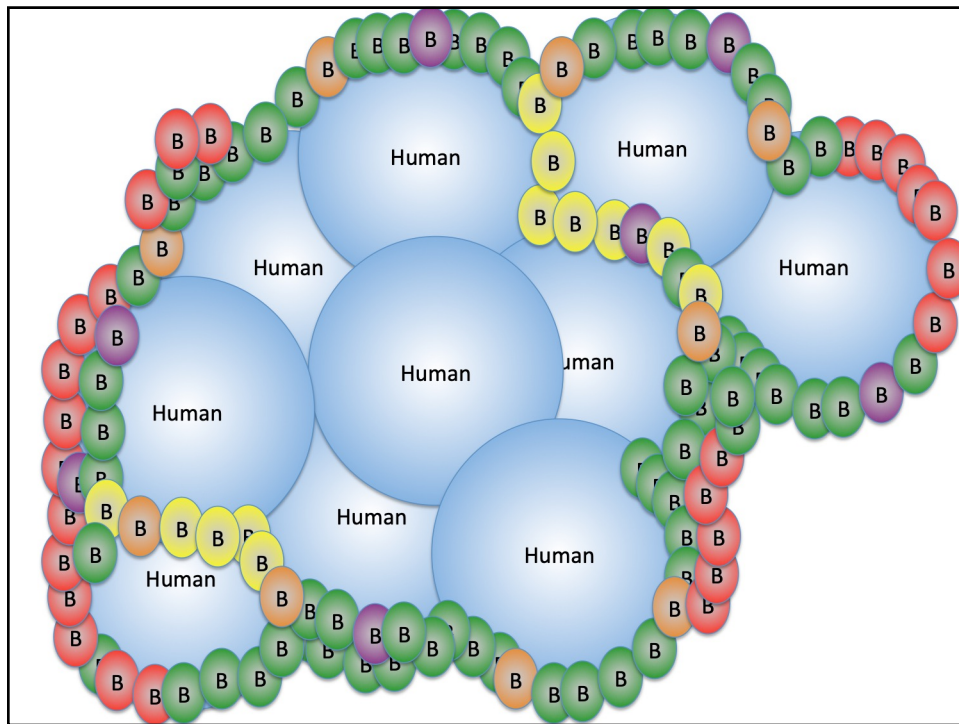
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Example: The Microbiome and Obesity



Walker, A. W. & Parkhill, J. *Science*. 2013

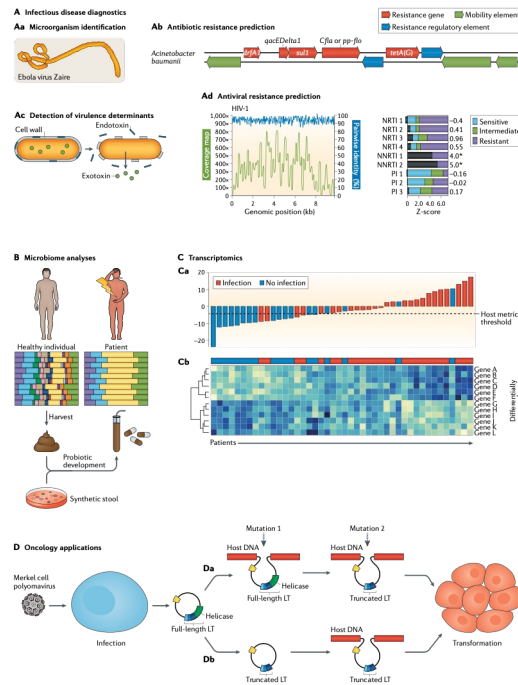
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Clinical Metagenomics (applications)

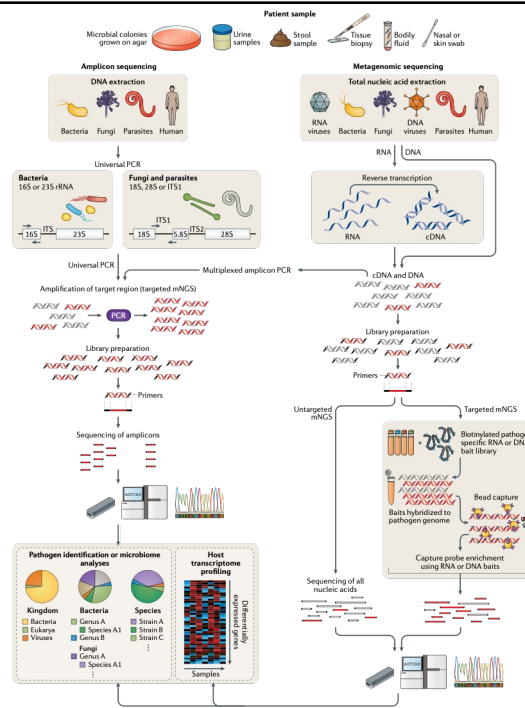
Chiu & Miller, Nat. Rev.
Gen, 2019



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Clinical Metagenomics (Methods)

Chiu & Miller, Nat. Rev.
Gen, 2019



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Shotgun Metagenomics

- Sequencing of DNA from all genomes in one sample (host and bacteria)
- Needs high coverage
- Challenges in assembly

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Challenges in Metagenomics

- Efficient DNA Isolation
- Biases in library prep and amplification
- Low coverage of 1000s of genomes
- Challenges in assembly



Complex Microbial Communities
(Real World)

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